
Numerical methods and optimization

Names of teacher: PhD Matuesz Gabor

Project Report: Automatic PID Controller Tuning using Numerical Optimization Methods

Author Name, index no., department	Adrian Ratajczak, 275533, W12N
	Bartosz Piotrowski, 275534, W12N
Submission date	15.06.2026



Politechnika
Wrocławska

1. Introduction

Proportional-Integral-Derivative (PID) controllers remain the industry standard for process control due to their simplicity and reliability. However, tuning these controllers for complex systems-especially those featuring significant thermal inertia and transport delays-can be highly challenging using traditional manual trial-and-error methods.

This project explores the application of numerical optimization techniques to automate the PID tuning process. By shifting the problem from manual heuristics to mathematical optimization, we can precisely target specific performance metrics like minimizing error and suppressing overshoot.

2. Project Objective

The primary objective of this project is to implement, evaluate, and compare two distinct numerical optimization algorithms for auto-tuning a PID controller:

- **Nelder-Mead Simplex Method** (a local, derivative-free direct search method implemented via MATLAB's `fminsearch`).
- **Particle Swarm Optimization (PSO)** (a global, heuristic stochastic algorithm implemented via MATLAB's `particleswarm`).

Both methods aim to minimize a custom objective function mapping to a first-order thermal plant with a time delay.

3. Project Description

3.1. Plant Model

The controlled system is a first-order thermal plant characterized by a static gain, a dominant time constant, and a transport delay. The ideal transfer function is defined as:

$$G(s) = \frac{K}{\tau s + 1} e^{-T_d s}$$

Where the system parameters are configured as:

- Static gain: $K = 2 \text{ C/W}$
- Time constant: $\tau = 60\text{s}$
- Transport delay: $T_d = 5\text{s}$

To simulate this within a linear system framework in MATLAB, the transport delay is approximated using a 3rd-order **Padé approximation**.

3.2. Optimization Metric (Cost Function)

The algorithms evaluate the controller settings based on a custom cost function J . The goal is to minimize the Integral of Squared Error (ISE) while heavily penalizing any overshoot to ensure system stability and material safety:

$$J = \int_0^{T_{sim}} e^2(t) dt + 500 * overshoot^2$$

Where $e(t) = r - y$ represents the tracking error given $r = 25^\circ C$ over a simulation time $T_{sim} = 400$ seconds.

3.3. Algorithm Configurations

- **Nelder-Mead (`fminsearch`):** Since it lacks built-in bound constraints, an auxiliary array q was introduced where the actual PID gains are calculated as $p = \exp(q)$. This mathematical transformation strictly guarantees that the gains remain strictly positive ($Kp, Ki, Kd > 0$).
- **PSO (`particleswarm`):** Configured with a swarm size of 40 particles and bounded by explicit lower boundaries ($[0.01, 0.001, 0]$) and upper boundaries ($[50, 5, 30]$).

4. Results and Analysis

The numerical results obtained from the simulations are summarized in the table below:

Tuning Method	Kp	Ki	Kd	Cost Function (J)
Manual Settings	1.000	0.0200	2.000	10137.65
Nelder-Mead	4.797	0.0965	14.665	3338.04
PSO	4.797	0.0965	14.665	3338.04

4.1. Simulation Visualizations

Below is the placeholder for the step response comparison chart generated by the script.

[INSERT GENERATED MATLAB PLOT HERE] > The plot should display the step responses for the Manual, Nelder-Mead, and PSO methods reaching the $25^\circ C$ setpoint, alongside the $\pm 2\%$ settling band.

4.2. Convergence Behavior

- **Nelder-Mead:** Terminated successfully after 79 iterations (149 function counts), converging rapidly due to a well-chosen initial guess.
- **PSO:** Successfully navigated away from an initially unstable region ($J = 10^8$) and found the global minimum within 50 iterations, terminating after detecting a stall in objective value changes.

5. Conclusion

The project successfully demonstrates the power of numerical optimization over manual tuning. Both the Nelder-Mead algorithm and **Particle Swarm Optimization** converged to the exact same optimal controller gains ($K_p \approx 4.797, K_p \approx 0.0965, K_p \approx 0.0965$).

This identical convergence indicates that the optimization landscape possesses a distinct, well-defined global minimum. By utilizing these algorithms, the cost function value was reduced by approximately 67% compared to the manual baseline, yielding a faster, highly optimized step response without compromising the system with excessive overshoot.